Fe-C Problem 4 | Problems T4_PhaseDiagrams II.pdf #4 | Fe-C · ledeburite

Phases and Microconstituents of a Eutectic Cast Iron

Original statement · Problems T4_PhaseDiagrams II.pdf, p. 2

4.- For an alloy Fe-4.3% of C, at the following temperatures: T_1 = 1147 + ΔT ºC; T_2 = 1147 – ΔT ºC; T_3 = 900 ºC; T_4 = 727 + ΔT ºC; T_5 = 727 – ΔT ºC. a) Indicate the phases present, their chemical composition and their mass fractions. b) Indicate the micro constituents, their chemical composition and their mass fractions.

Official Solution (the phase symbols print as empty boxes in the PDF and are restored from context): a) T_1: C_L = 4.3% C, w_L = 1; T_2: C_{Fe_3C} = 6.7% C, C_γ = 2.14% C, w_γ = 0.53; T_3: C_γ = 1.3% C, C_{Fe_3C} = 6.7% C, w_γ = 0.44; T_4: C_γ = 0.76% C, C_{Fe_3C} = 6.7% C, w_γ = 0.40; T_5: C_α = 0.022% C, C_{Fe_3C} = 6.7% C, w_α = 0.36; b) At T_2: $w_{Fe_3C,E}$ 0.47, $w_{\gamma,E}$ 0.53

PHASE 1 · HYPOTHESES & PARAMETERS

1. Equilibrium cooling, Fe-Fe3C diagram [Session 9 Slide 4]; $C_0 = 4.3$ % C is the **eutectic composition**: the alloy is liquid down to 1147 °C and then becomes 100 % **ledeburite**, $L \rightarrow \gamma\,(2.14) + \mathrm{Fe_3C}\,(6.67)$ [Session 9 Slide 10].
2. **Reading of the figure:** A_{cm} at 900 °C ≈ 1.3 % C (as in Problem FC03).

PHASE 2 · FRAMES & STRATEGY

Frame: the Fe-Fe3C diagram (wt% C, °C); one tie line per temperature. Below 1147 °C the alloy stays in γ + Fe3C, and the γ of the ledeburite loses carbon along A_{cm} (more cementite precipitates), until at 727 °C it becomes pearlite ("transformed ledeburite", Session 9 Slide 11).

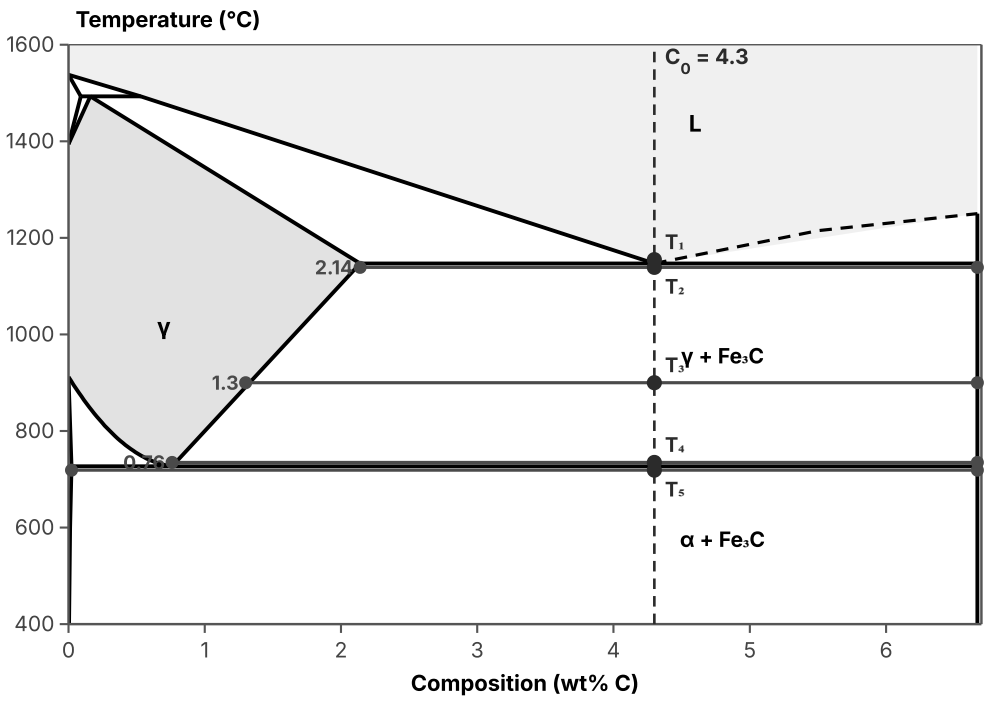


Figure 32. Fe-4.3 % C: liquid above 1147 °C, ledeburite (γ + Fe_3C) below it, and the tie lines at 900 °C and 727 ± ΔT.

PHASE 3 · STEP-BY-STEP DERIVATION

a) Phases T_1: liquid, 4.3 % C, $W_L = 1$.

T_2 (γ 2.14 / Fe3C 6.67; length 4.53):

$$W_\gamma = \frac{6.67 - 4.3}{4.53} = \frac{2.37}{4.53} = 0.523 \qquad W_{Fe_3C} = \frac{4.3 - 2.14}{4.53} = \frac{2.16}{4.53} = 0.477$$

T_3 (γ 1.3 / Fe3C 6.67; length 5.37):

$$W_\gamma = \frac{6.67 - 4.3}{5.37} = \frac{2.37}{5.37} = 0.441 \qquad W_{Fe_3C} = 1 - W_\gamma = 1 - 0.441 = 0.559$$

T_4 (γ 0.76 / Fe3C 6.67; length 5.91):

$$W_\gamma = \frac{6.67 - 4.3}{6.67 - 0.76} = \frac{2.37}{5.91} = 0.401 \qquad W_{\mathrm{Fe_3C}} = 1 - 0.401 = 0.599$$

T_5 (α 0.022 / Fe3C 6.67; length 6.648):

$$W_\alpha = \frac{6.67 - 4.3}{6.67 - 0.022} = \frac{2.37}{6.648} = 0.356 \qquad W_{\mathrm{Fe_3C}} = 1 - 0.356 = 0.644$$

b) Microconstituents

- T_1: liquid.
- T_2: 100 % **ledeburite** (4.3 % C overall), made of $W_{\gamma,E} = 0.523$ and $W_{\mathrm{Fe_3C},E} = 0.477$ (Worked example 14 of the theory).
- T_3, T_4: ledeburite whose austenite has rejected cementite: γ + Fe3C in the amounts of part (a).
- T_5: **transformed ledeburite**: the 0.401 of austenite at T_4 has become pearlite, inside a cementite matrix (0.599).

PHASE 4 · CHECKS & INTERPRETATION

- Mass balance at T_2: $0.523 \times 2.14 + 0.477 \times 6.67 = 1.12 + 3.18 = 4.30$ % C .
- The amount of cementite only grows on cooling (0.477 → 0.559 → 0.599 → 0.644), because the solubility of C in the iron phase only falls.
- **Limiting case** (fractions dimensionless, compositions in wt% C): at exactly 4.30 % C the primary fraction is 0 on both sides of the eutectic: the alloy is all ledeburite at T_2.

Check against the official solution

Agrees with the key at T_3, T_4 and T_5.

DISCREPANCY WITH THE OFFICIAL SOLUTION

At T_2 the key gives $w_\gamma = 0.53$. With 6.67 % C for cementite (the value of the slides), $w_\gamma = 2.37/4.53 = 0.523$ (0.52); the key's 0.53 follows from 6.7 % C, $2.4/4.56 = 0.526$. The same gap appears in part (b): the ledeburite is $W_{\gamma,E} = 0.523$ and $W_{\mathrm{Fe_3C},E} = 0.477$ (0.52/0.48), against the key's 0.53/0.47. The difference is only the rounding of the cementite composition (the sheet's diagram ends at 6.7 % C); 0.523 and 0.477 are kept.